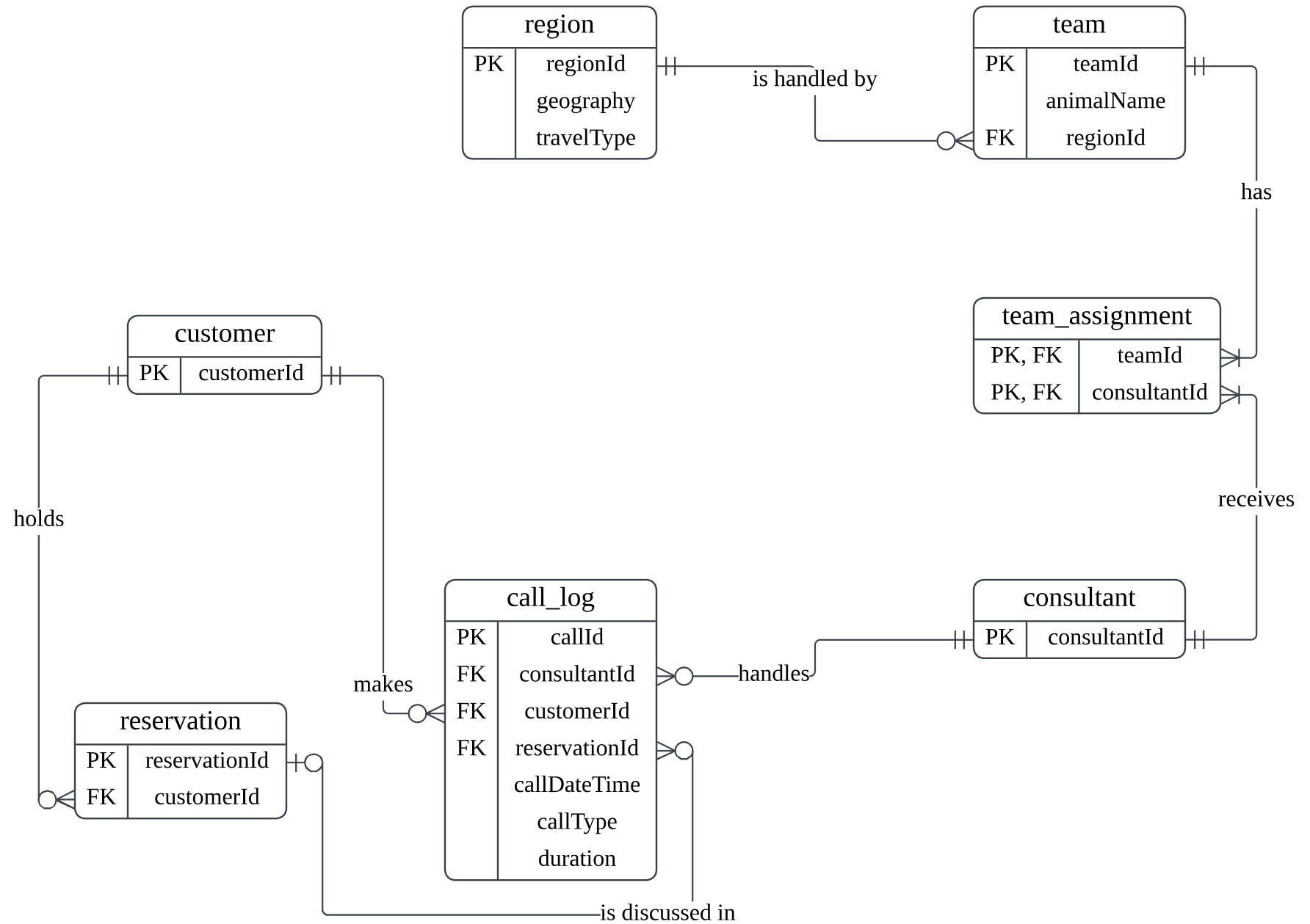


Exercise 1: Travel Consulting Company ER Model

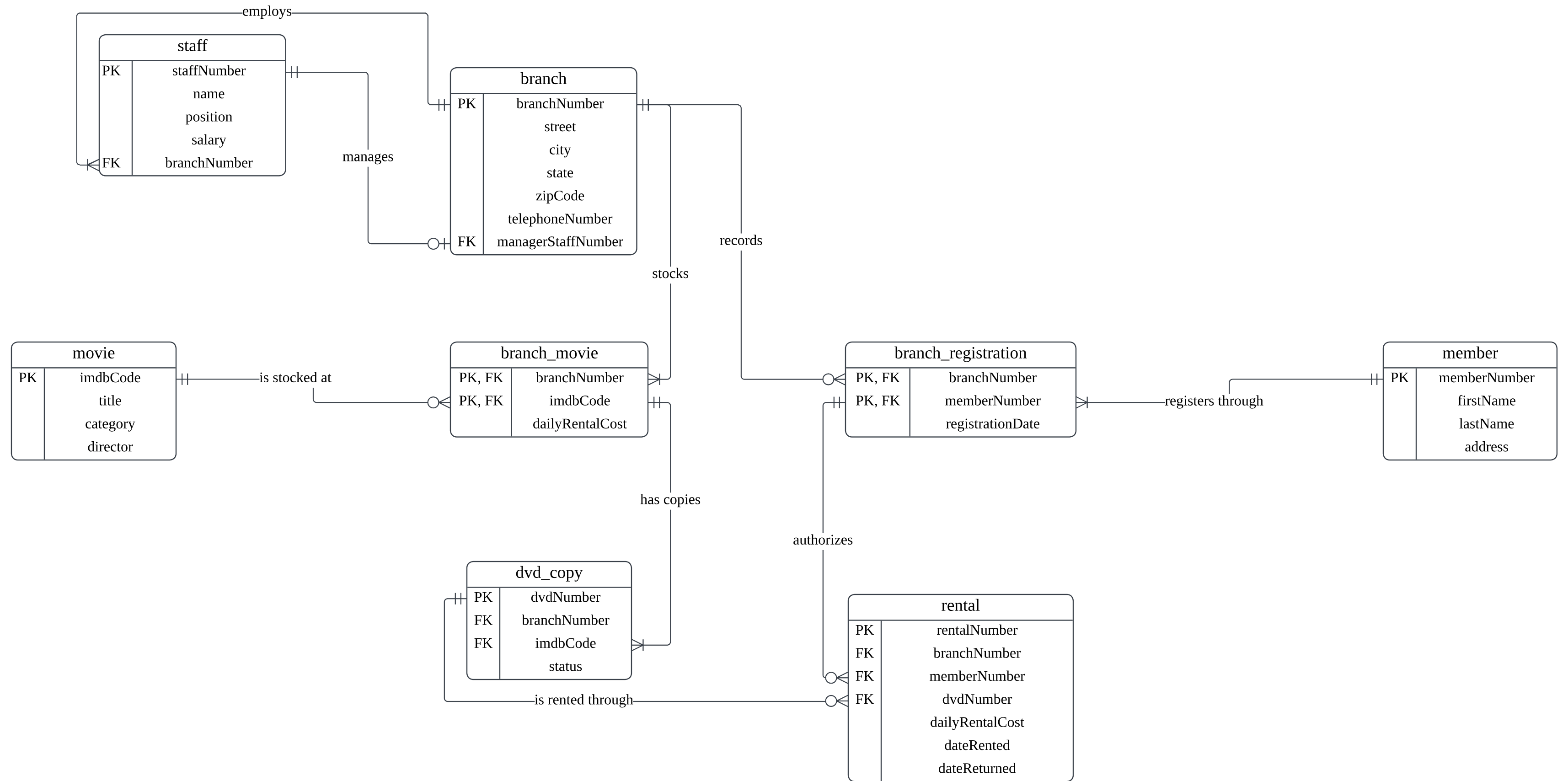


Exercise 1: Assumptions and Business Rules

1. The company has 12 teams. Each team serves exactly one region at a time. A region may be served by multiple teams or may temporarily have no assigned team. Only current team-region and consultant-team assignments are stored; assignment history is outside the scope of this model.
2. Each team has at least one consultant, and each consultant belongs to at least one team. The combination of teamId and consultantId uniquely identifies a team assignment.
3. Each region is described by its geography and one predominant travel type.
4. Each call is handled by exactly one consultant and is associated with exactly one customer. Customers include prospective travelers who have not yet made a reservation.
5. Each reservation belongs to exactly one customer. A call concerns either a new travel need or an existing reservation and refers to at most one reservation. reservationId is NULL for a new-travel-need call and is required for an existing-reservation call. Any referenced reservation must belong to the customer associated with the call.
6. callDateTime records when the call starts, and duration is measured in seconds. Monthly statistics are calculated from call_log by grouping calls by date and callType and summarizing their counts and durations; no separate statistics table is stored.

Legend: PK = primary key; FK = foreign key. Fields marked PK within the same table together form its primary key.

Exercise 2: DVD Rental Company ER Model



Exercise 2: Assumptions and Business Rules

1. Each staff member works at exactly one branch at a time. Only the current assignment is stored; reassignment history is outside the scope of this model. Each branch has at least one staff member and exactly one manager. The manager must be a staff member assigned to that same branch. managerStaffNumber is UNIQUE and NOT NULL.
2. Each movie has one category and one director. An operating branch stocks at least one movie, and each branch_movie record represents a movie for which that branch holds at least one DVD copy. Movie details may remain in the catalog even when no branch holds a copy.
3. The combination of branchNumber and imdbCode forms the primary key of branch_movie. In dvd_copy, (branchNumber, imdbCode) is one composite foreign key referencing branch_movie. The “has copies” relationship represents this entire pair, although the connector is attached at the imdbCode row. DVD copies are assumed to remain assigned to the same branch throughout their rental history.
4. memberNumber identifies a member across the company. Each member registers at one or more branches. The combination of branchNumber and memberNumber uniquely identifies a branch registration, and registrationDate records when that member registered at that branch.
5. Each rental record represents one DVD copy rented on one occasion. In rental, (branchNumber, memberNumber) is one composite foreign key referencing branch_registration. The “authorizes” relationship represents this entire pair, although the connector is attached at the memberNumber row. The rental branch must also match the branch holding the referenced DVD copy, and registration must occur no later than the rental date.
6. branch_movie.dailyRentalCost stores the current daily rate shared by all copies of that movie at that branch. rental.dailyRentalCost records the daily rate agreed when the rental begins and is retained when the current rate changes.
7. dateReturned is NULL until the DVD is returned and cannot precede dateRented. A DVD copy may have many historical rentals but at most one active rental at a time. Its status must be consistent with its rental availability.
8. The ten-DVD limit is interpreted as a company-wide maximum of ten simultaneous, unreturned rentals per member. It does not limit the number of historical rental records. This rule requires validation beyond the relationship cardinalities shown.
9. The member’s full name and the movie title associated with a rental are obtained through the related member, branch_registration, dvd_copy, branch_movie, and movie records. They are not duplicated in rental.

Legend: PK = primary key; FK = foreign key. Fields marked PK within the same table together form its primary key. A composite foreign key references the corresponding combination of fields as a unit.